

Creatine supplementation combined with blood flow restriction training enhances muscle thickness and performance: a randomized, placebo-controlled, and double-blind study.

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Creatine supplementation combined with blood flow restriction training enhances muscle thickness and performance: a randomized, placebo-controlled, and double-blind study.

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Abstract

This study aimed to compare the effects of an 8-week Creatine (Cr) or placebo (PL) supplementation on muscle strength, thickness, endurance, and body composition employing different training paradigms with blood flow restriction (BFR) vs. traditional resistance training (TRAD). Seventeen healthy males were randomized between the PL ($n = 9$) or Cr ($n = 8$) groups. Participants were trained unilaterally utilizing a within-between subject bicep curl exercise where each arm was allocated to TRAD or BFR for eight weeks. Muscular strength, thickness, endurance, and body composition were evaluated. Creatine supplementation promoted increases in muscle thickness in TRAD and BFR, compared to their placebo counterparts, however, without significant difference between treatments ($p=0.349$). TRAD training-induced increases in maximum strength (1RM) compared to BFR after eight weeks of training ($p = 0.021$). Repetitions to failure at 30% of 1RM were increased in the BFR-Cr group compared to TRAD-Cr ($p = 0.004$). Repetitions to failure at 70% 1RM were increased from weeks 0-4 ($p<0.05$) and 4-8 ($p<0.05$) in all groups. Creatine supplementation exerted a hypertrophic effect when utilized with TRAD and BFR paradigms and increased muscle performance at 30% 1RM when utilized in conjunction with BFR. Therefore, creatine supplementation seems to amplify muscle adaptation following a BFR program.

Registered in the Brazilian Registry of Clinical Trials (ReBEC), under the registration number: RBR-3vh8zgj.

Keywords: Creatine monohydrate, Nutritional supplementation, Blood flow restriction, Hypertrophy, Strength, Muscular endurance.

55 Introduction

56
57 Increases in muscular strength and hypertrophy are important adaptations associated with
58 resistance training that can translate to clinically meaningful outcomes in various populations,
59 including athletes, the elderly, and the general population (Fry, 2004). The American College of
60 Sports Medicine recommends resistance training with intensities over 70% of 1 repetition
61 maximum (1RM) to optimize strength and hypertrophic adaptations (Ratamess, 2009). However,
62 several studies have demonstrated that low-load (30% 1RM) training with blood flow restriction
63 (BFR) is capable of producing similar improvements in muscular strength and hypertrophy as
64 traditional high-load resistance training (TRAD) (Farup et al., 2015; Kim et al., 2017). For
65 example, in a seminal study by Takarada et al. (2000), high-load training elicited increases in
66 muscle cross-sectional area (CSA) and strength similar to that observed with low-load BFR.
67 Notwithstanding, a frequent finding in the literature for low-load training is reduced gains in
68 maximal strength compared to high-load, which seems to be due to lower neural changes
69 (Duchateau et al., 2021).

70 Regarding the mechanisms involved in such a response, it is clear that BFR significantly
71 increases muscle protein synthesis and robustly activates the mTORC1 compared to a similar
72 exercise performed in a free-flow manner (Gundermann et al. 2012). Factors involved in such a
73 response are the accumulation of lactate, which has been shown to increase muscle hypertrophy
74 in muscle cells (Ohno et al. 2018), the accumulation of H⁺ ions resulting in increased muscle
75 fatigue and subsequent type II fiber recruitment, increased venous pooling leading to increased
76 cell swelling and possibly intracellular factors reinforcing protein synthesis (Cholewa et al. 2018;
77 Suga et al. 2012; Petrick et al. 2019).

78 The principles of exercise biochemistry indicate that exercises performed with high loads
79 heavily rely on phosphocreatine metabolism for energy production. In contrast, low-load
80 exercises heavily rely on glycolysis for energy production (Sahlin et al., 2014). Although this is
81 still a valid assumption, an exception occurs during low-load BFR exercises, where muscle
82 phosphocreatine is robustly consumed, indicating a shift in energy metabolism compared to free-

flow exercises, possibly indicating a mismatch between phosphocreatine usage and resynthesis (Harris et al., 1976). Regarding the latter, Suga et al. (2012) observed that during BFR sets, phosphocreatine is consumed (although to a lesser extent if compared to a TRAD exercise). Still, it is not restored during the rest interval between sets, mainly because of muscle hypoxia (Harris et al., 1976). The final result of usage and lack of resynthesis is a progressive reduction of muscle phosphocreatine levels through multiple sets with BFR, which may be involved with the decreased muscle force response observed at the end of a BFR protocol (Suga et al., 2012).

Resistance training practitioners have used creatine supplementation due to its ability to increase creatine and phosphocreatine levels (Harris et al., 1992), thus enhancing muscle strength and muscle mass levels (Branch, 2003). Since decreased phosphocreatine levels are related to fatigue during BFR exercises (Suga et al., 2012), creatine supplementation would be an interesting alternative for enhancing muscle adaptations following a BFR program. Only one study investigated the interactions between BFR and creatine supplementation to the best of our knowledge. In this study, Medrano et al. (2019) did not observe any additional effect of creatine supplementation on BFR regarding increasing maximal isometric voluntary strength or muscle thickness in the elbow flexors. From this study, some important methodological questions arise: 1) the lack of specificity of the isometric contractions to evaluate the effects of dynamic training; and 2) the short-term nature of this study (3 weeks), which seems more appropriate to study neural adaptations, given that muscle hypertrophy is a relatively slow process (Damas et al., 2016).

Thus, the primary purpose of this study was to compare, in a time-course and double-blind placebo-controlled study, the effects of 8 weeks of creatine supplementation during BFR (30% 1RM) or TRAD (70% 1RM) on dynamic muscle strength and muscle thickness. A within-subject design and unilateral elbow flexion resistance training program was employed. Volume load of training (VLT [i.e., Sets x Number of Repetitions x Load]), 1 Repetition Maximum, repetitions to failure at 30% and 70% of 1RM, muscle thickness, and body composition were evaluated at baseline, four weeks, and eight weeks. We hypothesized that creatine supplementation would improve all supra-cited muscle performance variables in both resistance training paradigms compared to placebo controls.

Methods

Experimental Approach to the Problem

This study used a randomized, double-blind design. Those not directly involved in the data collection generated the random allocation sequence to ensure blinding. Participants were pair-matched based on initial arm circumference and strength levels and then randomly assigned to one of two treatment groups: Placebo (PL) or Creatine (CR). Each arm was randomly assigned to either a blood flow restriction (BFR) or traditional (TRAD) resistance training condition. Participants were assessed for anthropometric measures and arterial occlusion pressure (AOP) on the first visit. During visit two, participants performed an elbow flexion 1 RM; during visits four and five, the repetitions to failure test with 30% and 70% of 1RM were performed within 48 hours between tests. Repetitions to failure, strength tests, muscle thickness, and body composition were also performed after four and eight weeks of the intervention. Training occurred for eight weeks, with each arm being trained twice weekly on separate days, and each session separated by 72 hours. The total number of repetitions performed and load lifted was recorded for each set, and the VLT was calculated.

Insert Figure 1

Subjects

Of 61 men who participated in the first screening, 24 met all the inclusion criteria and agreed to participate in the study protocol. Participants were pair-matched based on initial arm circumference and strength levels and then randomly assigned to one of two treatment groups: Creatine (n=12) and Placebo (n=12). Four participants from Creatine and three from the placebo group the intervention dropped out of the study for unspecified reasons. Thus, the final sample analyzed was eight subjects in the Creatine group and nine in the Placebo group. Subjects were non-vegetarian, non-vegan, untrained male graduate students at the Federal University of Maranhão were recruited for this study, which was approved by the Research Ethics Committee

(CAAE: 91094318.7.0000.5087), and all research was conducted according to the 2013 Revision of the Declaration of Helsinki. It was properly registered in the Brazilian Registry of Clinical Trials (ReBEC) under the registration number: RBR-3vh8zgj. Participant characteristics are presented in **Table 1**. All subjects read and signed an informed consent form. The inclusion criteria for the study were: 1) Between 18 to 30 years of age, 2) had not participated in regimented resistance training in the previous six months, 3) had not used any dietary supplement or ergogenic drug for at least two years before the study and 4) did not smoke or drink alcohol within proximity (at least one week) of testing visits.

Insert table 1

Insert Figure 2

Procedures

Anthropometric Measurements, Body Composition, and Dietary Intake Assessments

Body mass was measured using an electronic scale (Filizzola PL 50, Filizzola Ltda., Brazil) with a precision of 0.1 kg. Height was measured on a fixed stadiometer (Sanny, São Paulo, Brazil) with an accuracy of 0.1 cm and a length of 2.20 m. The arm circumference was measured with a relaxed arm along the body, at a distance two-thirds distal to the acromial and olecranon at the largest portion of the volunteer's biceps. From the results obtained, the average error was estimated as follows: 0.04 cm for arm circumference, and the intraclass correlation was 0.98 to 0.99.

The body composition assessment was done using the Bodymetrix Pro System (Intelametrix Inc., Livermore, CA, USA) by a single trained ultrasound technician according to the manufacturer's user manual and procedures adopted (Schoenfeld et al., 2019). Total fat-free mass (FFM) was then estimated by subtracting fat mass from body mass (Total body mass minus total fat mass = FFM), utilizing the Jackson & Pollock three folds protocol (thigh, chest, and abdominal).

All participants recorded food intake via a three-day food diary consisting of one weekend and two weekdays. All food intakes were analyzed for total kilocalorie, and macronutrient intakes

167 averaged for the three days at baseline, after four weeks, and after eight weeks of the beginning
168 of this study. Food diaries were analyzed using NutWin version 1.5 [Nutrition Support Program,
169 Federal University of São Paulo, São Paulo, Brazil, 2002].

170 **Determination of Arterial Occlusion Pressure** 171

172 The cuff used to determine the total occlusion pressure was the 6 cm wide (Nylon) model
173 (JPJ, Brazil). For the determination of Arterial occlusion of pressure (AOP), the cuff was placed
174 near the glenohumeral joint. The pulse was initially detected using a portable Doppler probe
175 (MEDMEGA, DV 610V), positioned on the radial artery, and subsequently inflated to 50 mmHg
176 with a manual sphygmomanometer (MISSOURI, Brazil) following the same protocol as
177 Loenneke et al., (2012). These procedures' re-test intraclass correlation coefficient proved reliable
178 (ICC= 0.91).

179 **Muscle Thickness** 180

181 A single-trained ultrasound technician obtained muscle thickness images of the elbow
182 flexor using an A-mode ultrasound imaging unit (Bodymetrix Pro System; Intelametrix Inc.,
183 Livermore, CA, United States). Following applying a water-soluble transmission gel (Mercur
184 S.A. - Body Care, Santa Cruz do Sul, Brazil) to the measurement site, a 2.5-MHz linear probe
185 was positioned perpendicularly to the tissue interface without depressing the skin. All
186 measurements were performed at the same point corresponding to 70% of the distance between
187 the acromial process and the lateral epicondyle, respectively, as Kim et al. (2017) described. The
188 images recorded in the computer were analyzed using the software Body View Pro-Fit
189 (Bodymetrix Pro System; Intelametrix Inc., Livermore, CA, United States), and muscle thickness
190 dimensions were obtained by the same ultrasound technician measuring the distance from the
191 subcutaneous adipose tissue–muscle interface to the muscle–bone interface, as described by
192 Brigatto et al., 2019. To help ensure that swelling in the muscles from training did not obscure
193 results, images were obtained 48-72 hours before the commencement of the study and after the
194 final training session. These procedures' re-test intraclass correlation coefficient proved reliable
195 (ICC = 0.99).

Dynamic Strength Test

One maximum repetition strength test (1RM): A warm-up set of 10 repetitions was performed with an unloaded 2 kg dumbbell rod. To determine the 1RM value of unilateral elbow flexion, subjects followed the test protocol according to the study by Yasuda et al., (2015). During the test, participants had their heels and backs resting on the wall, and feet separated by shoulder width to ensure positional standardization.

Repetitions to failure test: Subjects performed one set to muscular failure of the unilateral elbow flexion exercise with a load corresponding to 30% and 70% of the 1RM (obtained during basal measurements), respectively, with a constant speed controlled by a metronome (2-s eccentric and 2-s concentric) and 48 hours between tests. The total number of repetitions performed was recorded.

Supplementation Protocol

Initially, participants ingested 20 g of creatine (Creatine monohydrate Powder, Integralmedica®, Embu-Guaçu, Brazil) or a placebo (Maltodextrin – Integralmedica®, Embu-Guaçu, Brazil) in 5 g doses four times per day (breakfast, lunch, snack, and dinner) for five days (Hultman et al., 1996). Next, each subject consumed either 5 g of Creatine or a placebo immediately post-exercise for the remaining eight weeks. On the non-training days (three days per week), participants took the supplement home and were instructed to consume it at the same time as their regular training. A nutritionist delivered the supplement upon the presentation of empty packs by the subjects, and this professional was not part of the research team. Both creatine and placebo were in powder form and mixed by the nutritionist with 1 g of strawberry flavor and 250 ml of water (Clight Juice Kraft Foods Brazil - 200 mL: energy 21 kJ, 5 kcal, sodium 4,7 mg) to produce a taste, color, texture, and appearance that were indistinguishable between creatine and placebo. No subject reported any side effects of creatine supplementation (i.e., diarrhea or cramps).

Training Protocol

Initially, the subjects were introduced to the laboratory, after which they were instructed on the exercise to be used in the study. They were then familiarized with the BFR and TRAD dumbbell biceps curl exercises. The training protocol was randomly selected for each subject's arm, each arm was trained twice weekly, and each separate protocol was performed on non-consecutive days (i.e., BFR: Monday and Thursday, and TRAD: Tuesday and Friday). Subjects were instructed to perform each repetition starting with a fully extended elbow, flexing at the elbow to full flexion, and then returning to full arm extension (May et al., 2018). Repetition time was controlled by a metronome, with 2 seconds for concentric and 2 seconds for eccentric. The training program consisted of two progressive 4-week phases: 1st to 4th week, for BFR, the subjects performed one set of 30 repetitions and two sets of 15 repetitions with 30% of 1RM and 30 seconds of rest; For TRAD, three sets of 10 to 12 repetitions with 70% of 1RM and 120 seconds of rest were performed. After four weeks, from the 5th to 8th week of training, the BFR arm performed one set of 30 repetitions and three sets of 15 repetitions with 30% of 1RM and 30 seconds of rest; and, for TRAD, four sets of 10 to 12 repetitions with 70% of 1RM and 120 seconds of rest. Following the 4th week, the 1RM was reassessed, and the load used in weeks 5 through 8 was calculated based on the updated 1RM. For BFR, the subjects were equipped with a cuff to be placed near the glenohumeral joint and then inflated to a pressure corresponding to 50% of the AOP of the arm blood flow. The occlusion pressure was maintained throughout the exercise session, including rest intervals, and released only at the end. For TRAD, the intensity was controlled by the zone of repetition, and subjects needed to complete the movement failure with no more than 12 and not less than 10 repetitions. For BFR, sets were conducted for concentric movement failure, and subjects failed at every set (Farup et al., 2015).

Statistical analysis

Using a partial eta-square (0.19) for fat-free mass in the present study, and an alpha value of 5% via GPower 3.1 software, a 70% power to our sample size was found by analyzing the difference between groups post hoc. The Shapiro-Wilk test was applied to analyze the sample

distribution. Independent samples t-tests were used to compare general characteristics. A 2 x 8 (group x time) repeated measures analysis of variance (RM ANOVA) with the Bonferroni adjustment for multiple comparisons was used to compare VLT in each week of training between groups. A 2 x 3 (group x time) RM ANOVA with the Bonferroni adjustment for multiple comparisons was used to compare placebo and creatine groups on the dietary intake and macronutrient distribution (baseline, after 4 and 8 weeks of training). A 2 x 5 (group x time) RM ANOVA with the Bonferroni adjustment for multiple comparisons was used to compare placebo and creatine groups on the body composition and total body mass. The three-way ANOVA was used to analyze muscle thickness and performance for the longitudinal analysis. The supplementation group (placebo vs. creatine) was included as the between-subjects factor, and time (baseline to week 8) and training condition (BFR vs. TRAD) were used as the within-subjects factors. The interactions analyzed were treatment group x time, treatment group x training condition, training condition x time, and treatment group x training condition x time. The homoscedasticity of the data was verified before RM ANOVA analysis; the sphericity was verified according to Mauchly's W test, and the Greenhouse-Geisser correction was used when necessary. The data were analyzed using Statsoft Statistic software (version 10). Data are presented as mean and standard deviation (SD) or percent change. The statistician was blinded to both groups during data analyses.

Results

Twenty subjects were initially recruited for this study, but two subjects decided to leave the study for personal reasons. There was an outlier in the creatine group who was underweight and was excluded from the final analysis (body mass index < 18.5 kg/m²). Thus, the final sample analyzed was eight subjects in the creatine group and 9 in the placebo group.

Table 1 shows the general characteristics of the sample herein studied. There were no significant differences between any variables at baseline. There were also no statistically significant differences between groups for dietary intake (kcal), carbohydrates, proteins, or fats

expressed as grams or relative to body mass at any time (**Table S1. Supplementary material**). The AOP for the placebo group was 141.1±9.3 mmHg and 143.3±15.8 mmHg for the creatine group ($p=0.721$). The pressure corresponding to 50% of the AOP used during the training program was 71±4.6 mmHg for Placebo and 72±7.9 mmHg for the creatine group.

Insert Table S1

Table 2 compares placebo and creatine groups on body composition and total body mass across time.

Insert Table 2

There was a significant group $\times$ time interaction for total body mass ($F= 5.793, p= 0.007, \eta^2=0.28$) with significantly higher total body mass after weeks 4 ($p=0.04$) and 8 ($p=0.06$) compared to baseline only in the creatine group but not for placebo.

For fat-free mass, there were significant group $\times$ time interactions ($F= 3.549, p= 0.041, \eta^2=0.19$) with significantly greater FFM at week 4 (2.5%) ($p=0.007$) and week 8 (2.5%) ($p=0.013$) in relation to baseline only in the creatine group but not for placebo at week 4 (0.3%) and week 8 (0.3%). There was no significant main effect of time, group, or interaction for fat mass ($F= 1.513, p= 0.237, \eta^2=0.09$) and fat mass percentage ($F= 0.584, p= 0.564, \eta^2=0.04$).

Figure 3 compares placebo and creatine groups on total weight lifted in each week of training (VLT).

Insert Figure 3

For total weight lifted in each week of training (VLT), there was a significant group $\times$ time interaction ($F= 3.074, p= 0.005, \eta^2=0.16$), training condition $\times$ time interaction ($F= 5.254, p<0.001, \eta^2=0.25$) and treatment group $\times$ training condition $\times$ time interactions ($F= 4.055, p= 0.001, \eta^2=0.20$). There was a significant increase in VLT only in week 2 for TRAD-CR and week

3 for BFR-CR, with significant increases also observed after weeks 5 and 8. Statistically significant increases for TRAD and BFR were observed only at week 5, with no significant difference in the VLT for BFR from weeks 5 through 8.

For the placebo group, TRAD showed greater VLT than BFR across time ($p < 0.05$); however, for the creatine group, there was significantly higher VLT in the TRAD-CR from week 1 to week 6 compared to BFR-CR ($p < 0.05$), but without significant differences between treatment groups at week 7 ($p = 0.117$) and week 8 ($p = 0.167$). Bonferroni's post hoc also showed greater VLT in the BFR-CR compared to BFR at week 8 ($p = 0.024$).

Figure 4 shows the comparison between the placebo and creatine groups on performance.

Insert Figure 4

For maximum strength (1RM), there was a significant increase across time ($F = 38.331$, $p < 0.001$, $\eta^2 = 0.72$) and training condition $\times$ time interactions ($F = 6.658$, $p = 0.004$, $\eta^2 = 0.31$) with significant increased at week-8 compared to baseline ($p < 0.001$) and week-4 ($p < 0.001$). There were significantly higher strength increases in the TRAD compared to BFR at week 8 ($p = 0.021$). However, no significant treatment group $\times$ training condition $\times$ time interactions were observed ($F = 1.333$, $p = 0.279$, $\eta^2 = 0.08$).

For repetitions to failure with 70% of 1RM, there was a significant increase across time ($F = 97.208$, $p < 0.001$, $\eta^2 = 0.87$) and treatment group $\times$ time interactions ($F = 7.009$, $p = 0.003$, $\eta^2 = 0.32$), with significant increases at week 4 and week 8 compared to baseline ($p < 0.05$), and week 8 about week 4 ($p < 0.001$) but post hoc did not show significant differences between treatment groups ($p > 0.05$).

For repetitions to failure with 30% of 1RM, there was a training condition $\times$ time interaction ($F = 4.671$, $p = 0.017$, $\eta^2 = 0.24$) with significant increases at week 4 and week 8 compared to baseline ($p < 0.05$) and week 8 in relation to week 4 ($p < 0.001$) and treatment group $\times$ training condition ($F = 5.237$, $p = 0.037$, $\eta^2 = 0.26$) but no treatment group $\times$ training condition $\times$ time

interaction ($F= 2.461$, $p= 0.102$, $\eta^2=0.14$) was observed. There were significantly greater repetitions in the BFR training compared to TRAD ($p= 0.009$) at week 8, and BFR-CR was higher than TRAD-CR ($p=0.004$).

Figure 5 compares placebo and creatine on muscle thickness (mm) across time.

Insert Figure 5

For muscle thickness (mm), there was a significant treatment group $\times$ time interaction ($F= 5.683$, $p=0.008$, $\eta^2= 0.27$), but no treatment group $\times$ training condition $\times$ time interactions were observed ($F= 1.089$, $p= 0.349$, $\eta^2= 0.07$). There was a significant increase at week 4 and week 8 compared to baseline ($p<0.05$) and week 8 about week 4 ($p<0.005$).

Discussion

The main objective of this study was to examine if creatine supplementation would benefit a low-load/high-repetition resistance training paradigm with BFR, compared to a traditional high-load/low-repetition resistance training paradigm regarding muscular adaptations. Our results demonstrated for the first time that eight weeks of creatine supplementation during BFR training enhanced the exercise volume load of training (VLT) compared to the BFR placebo. In some variables, such as repetitions to failure at 30% 1RM, the BFR creatine-supplemented group performed better than the BFR placebo, TRAD placebo, and the TRAD creatine-supplemented group. Regarding the muscle thickness response, creatine supplementation promoted increases in muscle thickness in both paradigms, TRAD and BFR, compared to their placebo counterparts. However, the increases were not statistically significant, which can be partially explained by the limited number of subjects. On the other hand, mean values from this study were in line with data from a previously published creatine study on TRAD (Candow et al. 2011), reinforcing the validity of such results. Our results suggest that creatine supplementation may be considered hypertrophic for both BFR and TRAD and ergogenic for the BFR under our experimental settings.

These synergistic improvements may have important implications in our model and possibly in different ecological contexts, such as rehabilitation programs, where BFR and creatine supplementation may be combined to augment muscular adaptations without heavy loading.

The efficacy of creatine supplementation in BFR training was tested against the TRAD paradigm as it represents the recommended training method for muscle hypertrophy by professional organizations such as the American College of Sports Medicine (Ratamess, 2009). It was observed that eight weeks of TRAD increased bicep muscle thickness by 9.5% (versus pre-exercise condition), which is similar to previous mean data from the literature (Hubal et al., 2005). When associated with a TRAD paradigm, creatine supplementation resulted in a 17% increase in muscle thickness after eight weeks, indicating the efficacy of creatine for hypertrophy purposes. This magnitude of change in the elbow flexors is similar to Candow et al. (2011), who reported increases in elbow flexor muscle thickness of 6.3% and 16.4% for placebo and creatine, respectively, following 6 weeks of training. These results indicate that the TRAD method of training and creatine supplementation employed in this study was sufficient to increase muscle hypertrophy, similar to the greater body of research (Kreider et al., 2017).

To our knowledge, Kim et al. (2017) published the first study comparing low-load BFR to higher-load TRAD using a unilateral within-subjects design. In that study, 8 weeks of BFR and TRAD resulted in similar increases in elbow flexor hypertrophy, also observed in our 8-week study. Specifically, we found muscle hypertrophy of 9.1%, which is close to the 11.5% increase in 6 weeks presented by Farup et al. (2015) following a BFR program. In addition, in the present study, BFR training with creatine supplementation led to a 12.7% increase in muscle thickness. In a study performed by Medrano et al. (2019), it was demonstrated that BFR in association with creatine supplementation (3 weeks) did not enhance either bicep muscle thickness or maximal voluntary isometric strength compared with BFR alone. However, it should be noted that in this study (Medrano et al., 2019), the restriction pressure was 30% of AOP, and the rest between sets was 60 seconds, making the exercise protocol milder than ours. Interestingly, in our study, BFR groups (placebo versus creatine) were equalized to the number of repetitions and sets, making the only difference between both groups, the weight lifted per exercise session. Given the relationship

between load and hypertrophy (Radaelli et al., 2015), we hypothesize that greater rates of progressive overload with creatine supplementation likely presented a larger stimulus for muscle protein accretion, thus leading to greater muscle hypertrophy. Still, another known effect of creatine supplementation is to reduce markers of protein breakdown and such effects may have positively influenced our hypertrophy results (Parise et al. 2001).

Interestingly, in the TRAD paradigm, muscle hypertrophy was augmented by creatine supplementation (vs TRAD placebo group), despite no differences in VLT between groups. A possible explanation for the lack of increased VLT in the TRAD creatine group may be related to the task time of dynamic contractions. In the present study, TRAD subjects performed 10-12 repetitions per set, within 4 seconds per repetition, totaling approximately 40 to 48 seconds spent in each set. According to a previous meta-analysis (Branch, 2003), creatine supplementation has the most pronounced effect on short-duration (< the 30s) high-intensity intermittent exercises. Thus, a possible explanation for the lack of creatine effects following this paradigm is that task time may have overreached the ideal range for the maximal performance benefits of creatine.

Since VLT was not enhanced by creatine in the TRAD group, mechanisms other than increased exercise volume may have contributed to the increased hypertrophy response. Creatine supplementation has been shown to increase muscle differentiation in cell cultures and increase satellite cell activation in a murine model of compensatory hypertrophy (Dangott et al., 2000; Deldicque et al., 2007). Creatine supplementation has also been shown to increase muscle IGF-1 mRNA expression (Deldicque et al., 2005) and decrease plasma leucine rate of appearance (an indirect marker of muscle protein breakdown) in humans (Parise et al. 2001). In addition, when associated with TRAD resistance training, creatine supplementation was shown to increase muscle mass and myofibrillar protein content (Willoughby et al., 2001), without affecting the ratio of skeletal muscle mass to intracellular fluid volume (Ribeiro et al. 2020). Altogether, physiological creatine effects, independently of enhanced muscle performance, should be taken into consideration since both, increased muscle thickness and increased lean mass (Table 2), but not VLT was observed in this paradigm and treatment.

Creatine supplementation is widely known to be a performance enhancer (Casey et al., 1996), mainly because of its properties to be transported and converted in the skeletal muscle tissue to phosphocreatine (Clark, 1997). At first glance, low-load training has little to do with phosphocreatine metabolism. However, as demonstrated previously during a multiple-set BFR session, phosphocreatine resynthesis is inhibited by blood flow restriction, resulting in the state known as metabolic freezing (Suga et al., 2012). Under this condition, phosphocreatine levels are not restored during the rest intervals between sets (as should occur in a low load free-flow exercise) and are utilized by the skeletal muscle during each set, therefore causing a progressive phosphocreatine decrement into a depletion state, which may be related with the decreased muscle force generated at the end of a BFR session (Suga et al., 2012). Thus, due to this specific aspect of BFR, we postulated that creatine supplementation could increase muscle performance and enhance muscle adaptations to this paradigm. To our results, when evaluated for a single set performed to the concentric failure at 30% of 1RM, the maximal number of repetitions performed in the BFR creatine group was significantly greater than BFR placebo or the TRAD groups at 8 weeks. These findings are in line with our initial hypothesis that increased phosphocreatine stores may be related to increased work capacity in the BFR paradigm, which may be also related to the increased muscle hypertrophy response.

Conclusion

The present study demonstrated multiple benefits of creatine supplementation in terms of muscle hypertrophy for both TRAD and BFR and muscle performance for BFR at 30% of 1RM and VLT. These specific synergistic improvements may have important implications not only in our model but also in rehabilitation or athletic programs, whereby BFR and creatine supplementation may be combined to augment muscular adaptations without heavy loading.

Limitations

Unilateral training has limitations mainly related to increases in muscle strength in the contralateral arm and this effect is more pronounced when this arm is kept untrained, which was

not the case in this investigation. As a positive factor, such an exercise model is known to decrease the variability between trained limbs, since factors such as nutrition, sleep, and supplementation, are delivered equally to both upper limbs (in our case). In this study, we employed a classical creatine supplementation schedule (saturation and maintenance), to promote the well-known benefits of creatine uptake and storage in the muscle. However, since we were not able to measure muscle or blood biochemical markers, the mechanisms involved in the results herein presented are not known. Future studies employing biochemical and/or molecular muscle fiber analysis are warranted to elucidate possible mechanisms of action on these specific and fascinating aspects of skeletal muscle and creatine metabolism.

Conflicts of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability Statement

The authors confirm that the data supporting the findings of this study are available within the article [and/or] its supplementary materials.

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Figures captions

Figure 1. Experimental Design.

Figure 2. Flow Chart

Figure 3. Comparison between placebo and creatine groups on total weight lifted in each week of training (VLT).

Figure 4. Comparison between placebo and creatine group on maximum strength and muscle endurance strength.

Note A= maximum strength 1RM B= muscle endurance with 70% of 1RM; C= muscle endurance with 30% of 1RM; a= main effect of time with significant difference between baseline ($p<0.05$); b= main effect of time with significant difference between week 4 ($p<0.001$). £= training condition $\times$ time interaction with higher values in the TRAD group in relation to BFR ($p= 0.021$); #= training condition $\times$ time interaction with higher values in the BFR group compared to TRAD ($p= 0.017$); \$= treatment group $\times$ training condition interaction with significant difference between training conditions within Creatine group ($p= 0.004$).

Figure 5. Mean muscle thickness values across time on placebo and Creatine group. Data are presented as mean $\pm$ SD.

Note: muscle thickness (mm); BFR= blood flow restriction; BFR-Creatine= blood flow restriction plus Creatine; TRAD= traditional resistance training; TRAD-Creatine= traditional resistance training plus Creatine. a= main effect of time with significant difference between baseline ($p<0.05$); b= main effect of time with significant difference between week 4 ($p<0.005$).

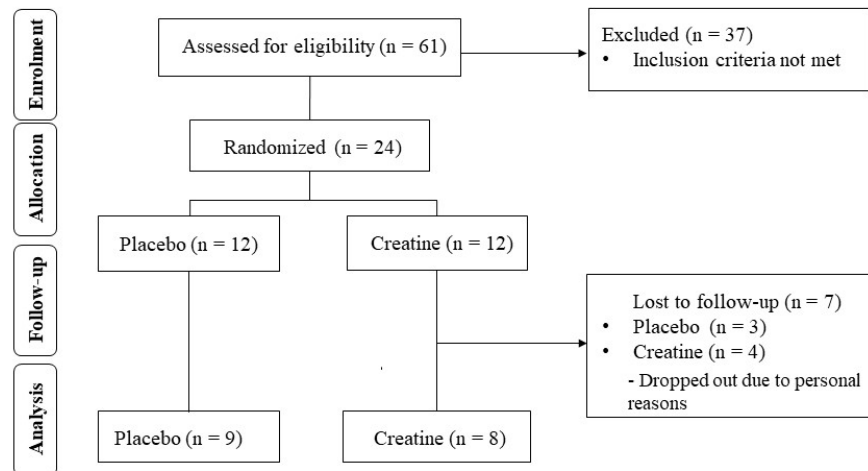


Figure 1 - Flow Chart

254x190mm (96 x 96 DPI)

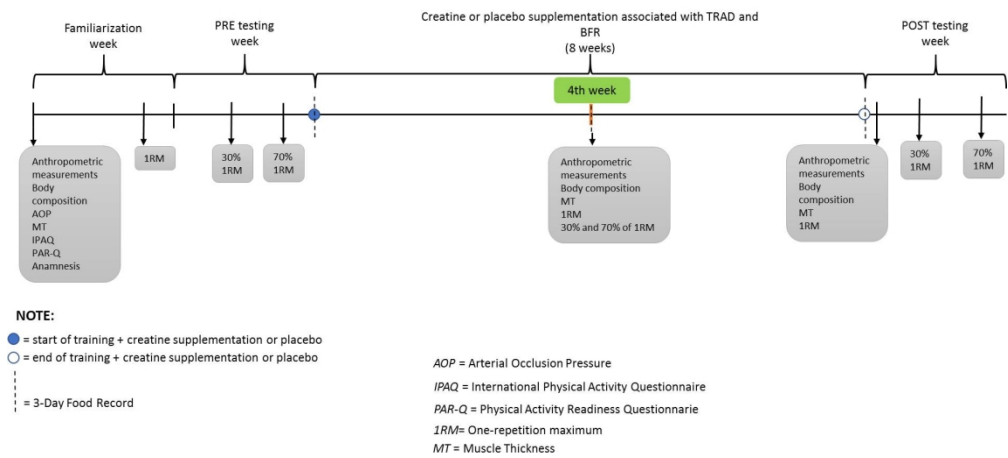


Figure 2. Experimental Design.
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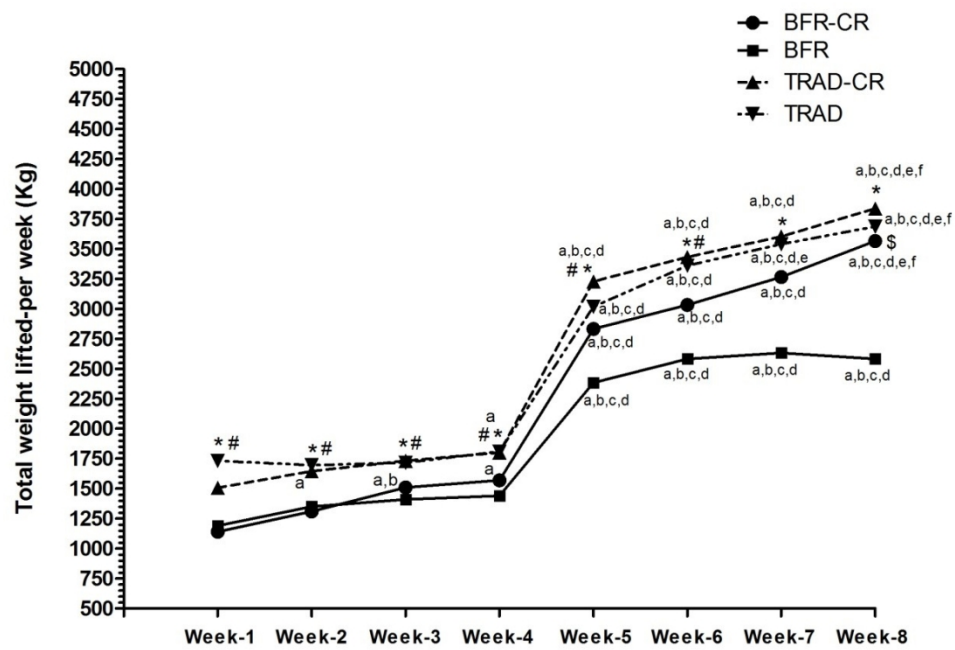


Figure 3. Comparison between placebo and creatine groups on total weight lifted in each week of training (VLT).

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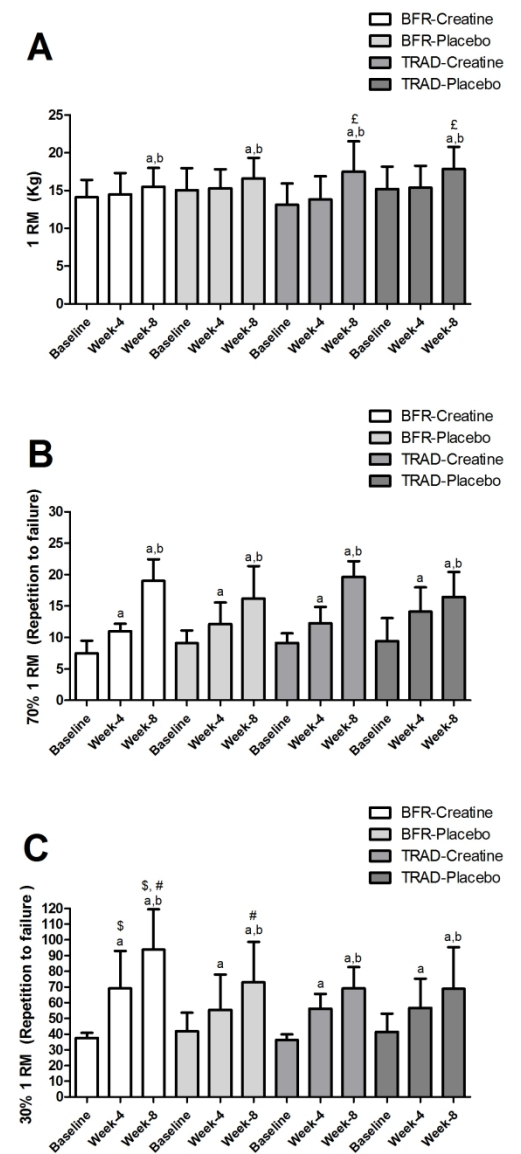


Figure 4. Comparison between placebo and creatine group on maximum strength and muscle endurance strength.

122x267mm (300 x 300 DPI)

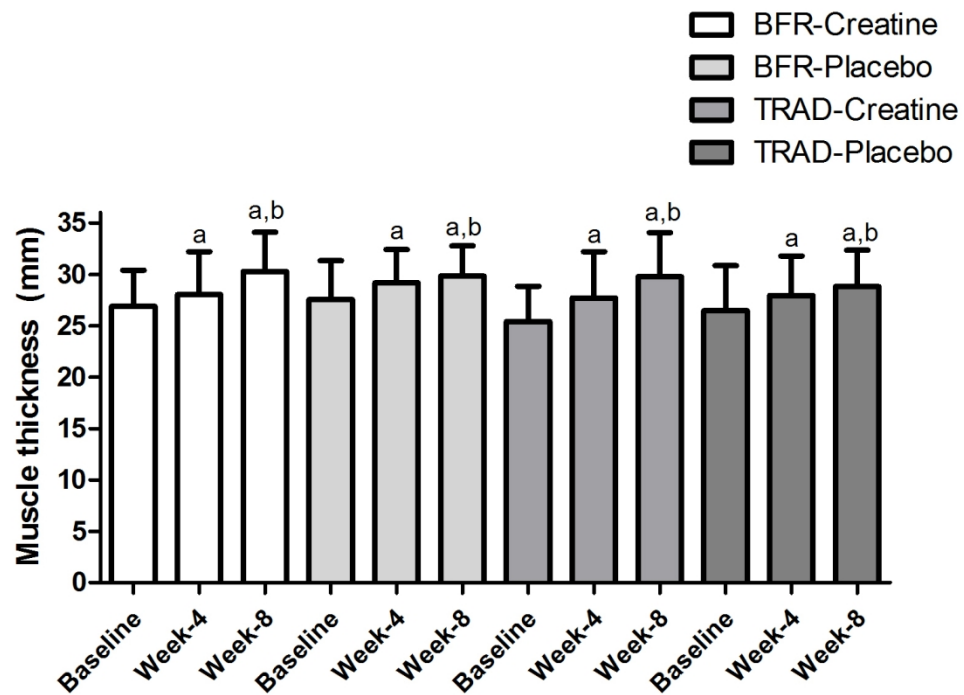


Figure 5. Mean muscle thickness values across time on placebo and Creatine group. Data are presented as mean $\pm$ SD.

137x101mm (300 x 300 DPI)

Table 1. General characteristics of the placebo and creatine groups.

Variable	Placebo (n=9)	Creatine (n=8)
Age (y)	20.6 ± 2.9	23.8 ± 3.7
Height (cm)	175.0 ± 0.1	172.0 ± 0.1
Weight (kg)	65.7 ± 8.9	70.0 ± 9.6

Note: Data are presented as mean +/- SD

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Table 2. Comparison between Placebo and Creatine groups on total body mass and body composition and across time.

Variables	Group	Baseline	Week-4	Week-8
Total body mass (kg)	CR	70.0±9.6	71.5±8.8 ^a	71.8±9.1 ^a
	PLA	65.7±8.9	65.8±9.4	65.7±8.8
FFM (kg)	CR	60.3±7.7	61.7±7.2 ^a	61.7±7.2 ^a
	PLA	57.8±6.9	58.1±7.5	58.0±7.2
Fat mass (kg)	CR	9.8±6.1	9.8±6.4	10.2±6.1
	PLA	7.8±3.8	7.7±3.5	7.8±3.3
Fat mass (%)	CR	13.6±7.1	13.3±7.5	13.8±7.1
	PLA	11.6±4.9	11.3±4.4	11.4±4.3

Note: CR= Creatine; PLA= Placebo; FFM= fat-free mass; a= significant difference from baseline.